

Active Directory Home Lab

Overview

This project is a basic Active Directory lab built to practice Windows Server administration and understand how domain environments work.

I used this lab to learn and practice Active Directory, Group Policies, and file sharing in a simple setup.

What I Practiced

Active Directory

- Created users and groups
- Organized users using Organizational Units (OUs)

Group Policy (GPO)

- Set password and account lockout policies
- Restricted Control Panel access
- Mapped network drives for users
- Blocked USB and removable storage
- Tested policies on client machines

File Sharing & Permissions

- Understood local vs network sharing

- Learned NTFS and share permissions
- Practiced read, write, and full control permissions
- Used inheritance and explicit permissions

File Server Resource Manager (FSRM)

- Set storage quotas
- Created file screening rules
- Set alerts for storage limits

Security Policies

- Applied account lockout policies
- Set different password policies using FGPP
- Restricted user logon access using GPO

Service Accounts

- Created a service account
- Set up automatic login for testing
- Made a simple kiosk-style setup with auto startup apps

What I Learned

- How Active Directory works in a domain
- How Group Policies affect users and computers
- How file permissions control access

- How organizations manage users and security

Tools Used

- Windows Server
- Active Directory Domain Services
- Group Policy Management
- Windows 10/11 Client
- File Server Resource Manager
- VirtualBox / VMware

Note

This is a learning project built in a virtual lab environment.